



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (AI&DS) (CBCS) Degree Examinations
 December 2023

AD1102-1 – FUNDAMENTALS OF DATA SCIENCE

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is data science primarily concerned with?**
 - A) Predicting the future
 - B) Analyzing and extracting insights from data
 - C) Writing code for web applications
 - D) Designing user interfaces
2. **Which of the following is NOT a common step in the data science workflow?**
 - A) Data collection
 - B) Data cleaning and preprocessing
 - C) Data visualization
 - D) Data encryption
3. **What is the first step in the data science process?**
 - A) Building predictive models
 - B) Data collection
 - C) Data visualization
 - D) Hypothesis testing
4. **Which phase of the data science process involves exploring and understanding the data to identify patterns and relationships?**
 - A) Data cleaning and preprocessing
 - B) Feature selection
 - C) Model training
 - D) Exploratory data analysis (EDA)
5. **Which of the following is an example of external data sources for data collection?**
 - A) Surveys and questionnaires
 - B) Customer transaction records
 - C) Social media posts
 - D) Data generated by experiments
6. **Which of the following is an example of qualitative data collection method?**
 - A) Conducting surveys with multiple-choice questions
 - B) Recording temperature readings every hour
 - C) Observing and recording customer reviews and sentiments
 - D) Counting the number of website visitors
7. **What is the primary goal of data preprocessing in the data science pipeline?**
 - A) To make the data smaller in size
 - B) To make the data more difficult to analyze
 - C) To prepare the data for analysis by addressing quality and structure issues
 - D) To randomly shuffle the data
8. **Which of the following is a common data preprocessing technique to handle missing values?**
 - A) Ignoring the missing values and proceeding with the analysis
 - B) Replacing missing values with random numbers
 - C) Imputing missing values with the mean or median of the available data
 - D) Deleting entire rows with missing values
9. **Which of the following is NOT a common technique used in Exploratory Data Analysis (EDA)?**
 - A) Creating scatter plots to visualize relationships between variables
 - B) Calculating summary statistics such as mean and standard deviation
 - C) Building complex machine learning models
 - D) Creating histograms to understand the distribution of data
10. **Skewness measures the:**
 - A) Spread or dispersion of data points in a distribution
 - B) Symmetry or asymmetry of data distribution
 - C) Peakness or flatness of a distribution
 - D) Range of values in a dataset
11. **Kurtosis quantifies:**
 - A) The presence of outliers in a dataset
 - B) The degree of skewness in a dataset
 - C) The peakedness or flatness of a distribution
 - D) The central tendency of a dataset

12. Which of the following is NOT considered a measure of central tendency in descriptive statistics?
 A) Mean B) Median
 C) Mode D) Variance
13. What does the MEAN represent in a dataset?
 A) The most frequent value B) The middle value when all values are sorted
 C) The average value of the dataset D) The value that occurs at the 75th percentile
14. Standard deviation measures:
 A) The range of values in a dataset B) The spread or dispersion of values in a dataset
 C) The total sum of all values in a dataset D) The frequency of each value in a dataset
15. In simple linear regression, what is the purpose of the regression equation?
 A) To predict the value of the dependent variable based on one independent variable. B) To calculate the variance of the dependent variable.
 C) To identify outliers in the data. D) To find the correlation coefficient between two variables.
16. What is the primary purpose of a residual plot in regression analysis?
 A) To visualize the distribution of the dependent variable B) To visualize the relationship between two independent variables
 C) To assess the linearity of the regression model and the randomness of residuals D) To compare the means of different groups in the dataset
17. How can you reduce generalization error in a machine learning model?
 A) By increasing the complexity of the model B) By using a smaller training dataset
 C) By using a larger testing dataset D) By regularizing the model and avoiding overfitting
18. What is the primary purpose of cross-validation in machine learning?
 A) To split the dataset into training and testing sets B) To ensure that the model is underfitting
 C) To evaluate a model's performance and estimate how it will perform on unseen data D) To increase the complexity of the model
19. What is the primary objective of using Ridge Regression in predictive modeling?
 A) To maximize model interpretability B) To minimize the bias in the model
 C) To reduce the variance in the model D) To increase the model's complexity
20. What is overfitting in the context of machine learning?
 A) It occurs when a model performs well on the training data but poorly on unseen data. B) It refers to a model that is too simple to capture the underlying patterns in the data.
 C) It describes a model that is just right and generalizes well to new data. D) It happens when the training data is too small.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

		Marks	BT*	CO*	PO*
1.	a) You are a student working on a data science project for your coursework, tasked with analyzing a dataset related to climate change. Apply the six key steps of the data science process to carry out the analysis.	8	L*3	1	1
	b) Illustrate the difference between BI & DS with regard to its characteristics.	8	L2	1	1
2.	a) You are given a Case Study: Fraud Detection and Prevention in a Financial Institution. In the context of Secure Bank's fraud detection and prevention efforts, discuss how the Five V's of Big Data is significant in the design and implementation of an effective fraud detection system.	8	L3	2	1
	b) Explain the evolution of Data Science with an example.	8	L2	1	1
3.	a) What is Null & Alternative hypothesis? Explain the steps involved in Hypothesis testing.	8	L2	2	1

- b) Consider a Confusion Matrix table:

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

Find:

- i) Accuracy
- ii) Error rate
- iii) Precision
- iv) F- measure

8 L3 2 1

Unit – II

4. a) Describe Exploratory Data Analysis and Explain the steps involved in data science 8 L2 3 1
- b) With a simple dataset explain how to calculate Skewness, Kurtosis and Anova in Python? 8 L3 3 1
5. a) Explain the assumptions underlying Multiple Regression. 8 L2 4 1
- b) The average rainfall (in mm) for the last 5 years in the state of Maharashtra was recorded along with the umbrellas sold in that particular year.
- i) Find the equation of linear regression.
 - ii) Suppose this year the state records a rainfall of 123.2 mm, what will be the expected number of umbrellas that will be sold?
 - iii) Calculate the coefficient of determination (R²) and comment on it.

Rainfall (mm)	Umbrellas Sold
121.2	52
152.6	72
98.4	40
171	100
85.6	34

8 L3 4 1

6. a) Implement the methods to calculate mean, median, standard deviation, variance and percentile for a data set using python. Mention the libraries that are commonly used for this purpose. 8 L2 3 1
- b) In the context of linear regression analysis, explain the significance of residual plots and distribution plots. How do these graphical tools help in assess the quality of a regression model and understand the distribution of variables in a dataset? 8 L2 4 1

Unit – III

7. a) Discuss the concepts of overfitting and underfitting in the context of machine learning and data science. Provide examples of when each of these issues may occur and explain the potential consequences of overfitting and underfitting on model performance. How can we address and mitigate these problems in their predictive modeling tasks? 8 L3 5 1
- b) Explain the process of using Grid Search for hyperparameter tuning in machine learning, particularly when dealing with models that have multiple hyperparameters. 8 L2 5 1
8. a) Explain the concept of cross-validation in machine learning and its importance in model assessment and selection. Compare and contrast three common types of cross-validation techniques. 8 L2 5 1
- b) Explain the concept of generalization error in the context of machine learning and its significance in model assessment and selection. Describe strategies and techniques that can be employed to mitigate generalization error. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
